2. db θ $a'a$、cc'，

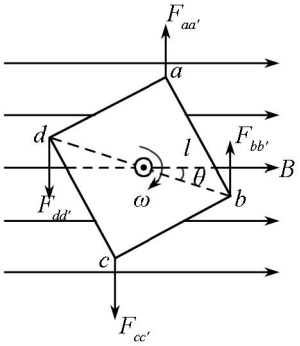
$b'b$、dd'

$$I_{a'a}=I_{ad}-I_{ba}=\frac{\sqrt{2}}{4}\frac{Bl^2\omega}{R}\sin\theta \qquad (10)$$

$$I_{cc'}=I_{bc}-I_{cd}=\frac{\sqrt{2}}{4}\frac{Bl^2\omega}{4}\sin\theta \qquad (11)$$

$$I_{b'b}=I_{ba}+I_{bc}=\frac{\sqrt{2}}{4}\frac{BI^2\omega}{R}\cos\theta \qquad (12)$$

$$I_{dd'}=I_{ad}+I_{cd}=\frac{\sqrt{2}}{4}\frac{Bl^2\omega}{R}\cos\theta \qquad (13)$$



$$F_{a'a}=F_{cc'}=BlI_{a'a}\,,\quad F_{b'b}=F_{dd'}=BlI_{b'b} \qquad (14)$$

$$M=2F_{a'a}\frac{\sqrt{2}}{2}l\sin\theta+2F_{b'b}\frac{\sqrt{2}}{2}l\cos\theta \qquad (15)$$

（15）（11）~（14）

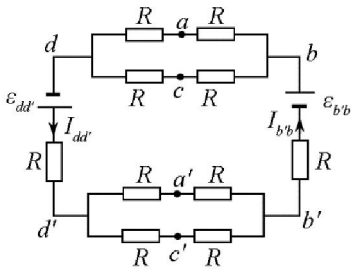
$$M=\frac{B^2l^4\omega}{2R}。 \qquad (16)$$

（16）（6）（4）（3）（2）

1（16）（15）（14）

2：

1. $a'a$ cc'



$$\varepsilon_{a'a}=\varepsilon_{cc'}=\frac{\sqrt{2}}{2}Bl^2\omega\sin\theta \qquad (1b)$$

$b'b$ dd'

$$\varepsilon_{b'b}=\varepsilon_{dd'}=\frac{\sqrt{2}}{2}Bl^2\omega\cos\theta \qquad (2b)$$

$\varepsilon_{a'a}$ $\varepsilon_{cc'}$ $\varepsilon_{b'b}$ $\varepsilon_{dd'}$ $a'a$ cc'

$b'b$ dd'

$$I_{b'b}^{(1)}=I_{dd'}^{(1)}=\frac{\varepsilon_{b'b}+\varepsilon_{dd'}}{4R}=\frac{\sqrt{2}}{4}\frac{Bl^2\omega}{R}\cos\theta \qquad (3b)$$

ba、ad、bc cd

$$I_{bd}^{(1)}=I_{ad}^{(1)}=I_{cd}^{(1)}=\frac{1}{2}I_{b'b}^{(1)}=\frac{\sqrt{2}}{8}\frac{Bl^2\omega}{R}\cos\theta \qquad (4b)$$

a'、a c、c'